

ADVANCED NEONATAL RESUSCITATION PROGRAMME

Study Notes and Procedure Checklist

Prepared for recertification exam: 14 January 2027

UHG Simulation Suite, University Hospital Galway

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For personal revision only. Does not replace the NRP Provider manual or UHG policy.

SECTION 1: INITIAL ASSESSMENT AT BIRTH

The first 60 seconds after birth are critical. At every birth, ask and act on three questions:

1. Is the baby term?
2. Does the baby have good muscle tone?
3. Is the baby crying or breathing?

If all three are yes, the baby stays with the mother for routine care. If any answer is no, move to resuscitation steps immediately.

Initial steps (first 30 seconds):

Warm and maintain normal temperature. Prevent heat loss. Avoid hyperthermia.

Position and clear the airway (sniffing position). Suction at 80 to 100 mmHg only if obstructed. Do not routinely suction.

Stimulate. Dry vigorously. Rub back, soles of feet, or trunk.

Evaluate. Assess respirations and heart rate at 30 seconds.

SECTION 2: HEART RATE ASSESSMENT

Target: above 100 bpm within 60 seconds of birth.

Below 100 bpm: begin positive pressure ventilation (PPV).

Below 60 bpm after 30 seconds of effective PPV: begin chest compressions.

SECTION 3: POSITIVE PRESSURE VENTILATION

Equipment: T-piece resuscitator or self-inflating bag with mask, size 1 for term neonate.

Rate: 40 to 60 inflations per minute.

Pressure: 25 to 30 cm H₂O (term infant). Up to 30 to 40 cm H₂O if chest not rising.

Oxygen: Begin with 21 percent (air) for term infants at 35 weeks or more.

Ventilation corrective steps (MR SOPA):

M: Mask adjustment. R: Reposition airway. S: Suction mouth then nose. O: Open mouth with jaw thrust. P: Pressure increase. A: Airway alternative (laryngeal mask or intubation) if above steps fail.

SECTION 4: CHEST COMPRESSIONS

Indication: heart rate below 60 bpm after 30 seconds of effective PPV.

Technique: two-thumb encircling technique. Compress lower third of sternum to one third of anterior-posterior depth.

Rate: 3 compressions to 1 ventilation. 90 compressions plus 30 breaths per minute.

Increase oxygen to 100 percent when compressions begin. Reassess every 60 seconds.

SECTION 5: MEDICATIONS

Epinephrine: indicated if heart rate remains below 60 bpm after 30 seconds of compressions with coordinated PPV.

Dose: 0.01 to 0.03 mg/kg IV via umbilical venous catheter (preferred). Concentration: 1:10,000 (0.1 mg/mL). Repeat every 3 to 5 minutes if no response.

Volume expander: normal saline 10 mL/kg IV over 5 to 10 minutes if blood loss suspected.

SECTION 6: APGAR SCORE REFERENCE

Sign	0	1	2
Colour	Blue/pale all over	Blue extremities, pink body	Pink all over
Respiration	Absent	Weak, irregular	Strong cry
Heart rate	Absent	Below 100 bpm	Above 100 bpm
Muscle tone	Limp	Some flexion	Active flexion
Reflex response	No response	Grimace	Cry or cough

Score 7 to 10: Normal. Score 4 to 6: Moderate depression. Score 0 to 3: Severe depression, immediate resuscitation required.

EXAM PREP CHECKLIST

Item	Status
Initial assessment steps	Reviewed
PPV technique and MR SOPA	Reviewed
Chest compression technique and rate	Reviewed
Epinephrine dose and route	Reviewed
UVC insertion	To review again 10 January 2027
Apgar scoring	Reviewed
Practice simulation booking at UHG	To book, target December 2026
NRP 8th edition updates	In progress
Exam date confirmed	14 January 2027, 9.00 AM, UHG Simulation Suite